第三章　分子的对称性和群论基础

- 对称操作和对称元素怎样数学表达？
- 什么是群？
- 什么是特征标？怎样用特征标表达对称性？
- 怎么利用特征标表理解红外光谱？

群论在化学中应用的核心很简单：
判断积分是否为零
高级版的“奇函数积分为零，偶函数不为零”

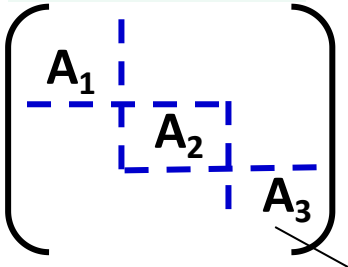
特征标表：科学版　以C_{3v}群为例

不同基组

C_{3v}	E	$2C_3$	$3\sigma_v$	基
A_1	1	1	1	z, z^2
A_2	1	1	−1	
E	2	−1	0	$(x, y), (xz, yz)$

i

具体矩阵



$\Gamma_i(R)$

简并度

表示矩阵的迹

$\Gamma_i(R)_{mn}$

$$\sum_i l_i^2 = h$$

$$\sum_R [\Gamma_i(R)_{mn}][\Gamma_j(R)_{m'n'}]^* = \frac{h}{\sqrt{l_i l_j}} \delta_{ij} \delta_{mm'} \delta_{nn'}$$

不同对称性的名字

说人话青春版

以C_{3v}群为例

C_{3v}	E	$2C_3$	$3\sigma_v$	基
A_1	1	1	1	z, z^2
A_2	1	1	−1	
E	2	−1	0	$(x, y), (xz, yz)$

对称元素

对称元素有几个

你看不懂但有用的数字

满足对称性的函数举例

请能欣赏这一页的读者直接跳到ppt第12页，以免引起不适

1. Characters of different irreducible representations are orthogonal:

The vectors whose components are the characters of two different irreducible representations are orthogonal, that is

$$\sum_R \chi_i(R)\cdot \chi_j(R) = 0 \quad when\ i \neq j$$

$$= \text{h} \quad \text{when i=j}$$

广义正交定理的三个推论

2. # of classes = # of irreducible representatives

3. $\sum_i l_i^2 = h$

h is The order of the group (the number of operations)

l_i is The dimension of the ith representation

第五节、特征标表

目标：描述一个对象在对称操作下变换的对称性

Character Tables

C_{3v}	E	$2C_2$	$3\sigma_v$	基	
A_1	1	1	1	z	x^2+y^2, z^2
A_2	1	1	−1	R_z	
E	2	−1	0	(x, y), (R_x,R_y)	(x^2-y^2,xy) (xz, yz)
II		I		III	IV

Area I : characters of the irreducible representations

Area II : Mulliken symbols

1) 1-D representation A [symmetric $\chi(Cn)=1$],
or B [antisymmetric $\chi(Cn)=-1$]

2) 2-D representation E / 3-D representation T

3) A_1 / A_2 and B_1 /B_2 refers to symmetric (1) and anti-symmetric (2) with respect to a C_2 perpendicular to the principal axis
If a C_2 axis is lacking, the subscript is with respect to a vertical plane of symmetry

4) primes and double primes indicates symmetric and anti-symmetric with respect to σ_h

5) g -- symmetric with respect to a center of inversion
u -- antisymmetric with respect to a center of inversion

Area III : always have six symbols: x,y,z , R_x, R_y, R_z

x, y, z -- coordinates

R_x, R_y, R_z – rotations about the axes specified in the subscript

Area IV : listing all of the squares and binary products of coordinates according to their transformation properties.

把可约表示转化为不可约表示

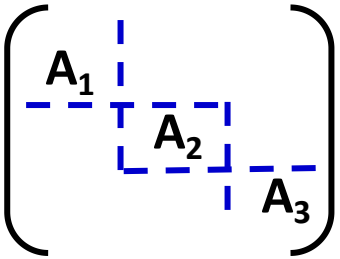
The original big matrix ⟶ Reducible representation

↓ similarity transformations

Block diagonalized matrix

↓

不可约表示



A_1, A_2, A_3 … cannot be further reduced into smaller matrices

A_1, A_2, A_3 … are **irreducible representations**.

The irreducible representations are the simplest representations of the operations of the point group.

Decomposition of reducible representations

分解可约表示为不可约表示

C_{2v}	E	C_2	σ_v	$\sigma_{v'}$
A_1	1	1	1	1
A_2	1	1	-1	-1
B_1	1	-1	1	-1
B_2	1	-1	-1	1
Γ_{red}	3	1	3	1
$2A_1$	2	2	2	2
B_1	1	-1	1	-1

Sum=Γ_{red}

So $\Gamma_{red} = 2A_1+B_1$

How about more complicated examples ?

约化公式

- $a_i = \frac{1}{h}\sum_R (\chi^R \cdot \chi_i^R \cdot C^R)$
- a_i= the number of times the irreversible representation Γ_i appears in the reducible representation Γ_{red}
- h = order of the point group
- R = an operation of the group
- χ^R = character of the operation R in the reducible representation Γ_{red}
- χ_i^R= character of the operation R in the irreversible representation Γ_i
- C^R= the number of members in the class to which R belongs

C_{3v}	E	$2C_2$	$3\sigma_v$
A_1	1	1	1
A_2	1	1	–1
E	2	–1	0
Γ_{red}	12	0	2

$a_{A1}= (1/6)[12*1*1+0*1*2+2*1*3]=3$

$a_{A2}= (1/6)[12*1*1+0*1*2+2*(-1)*3]=1$

$a_{E}= (1/6)[12*2*1+0*(-1)*2+2*0*3]=4$

$\Gamma_{red}=3A_1+A_2+4E$

？可约表示从哪来？

第六节、利用群论分析分子的振动

目标：分析红外振动峰的个数

- 简正振动的数目和对称类型
- 分子的简正振动数：
  - 每个原子的运动自由度：3
  - 分子（n个原子）自由度：3n.
    - 在这3n个自由度中
      - 三个自由度属于整个分子朝三度空间的三个方向，如以笛卡尔坐标表示，则为x、y和s方向作平移运动；
      - 三个自由度属于所有质子一齐绕x、y或z轴作转动运动.
  - 振动自由度：3n-6
    - 对于线型分子，由于分子只能绕垂直于键轴方向的两根轴中的任一根转动，而不能绕键轴本身转动，因此，线型分子具有(3n—5)个简正振动方式.

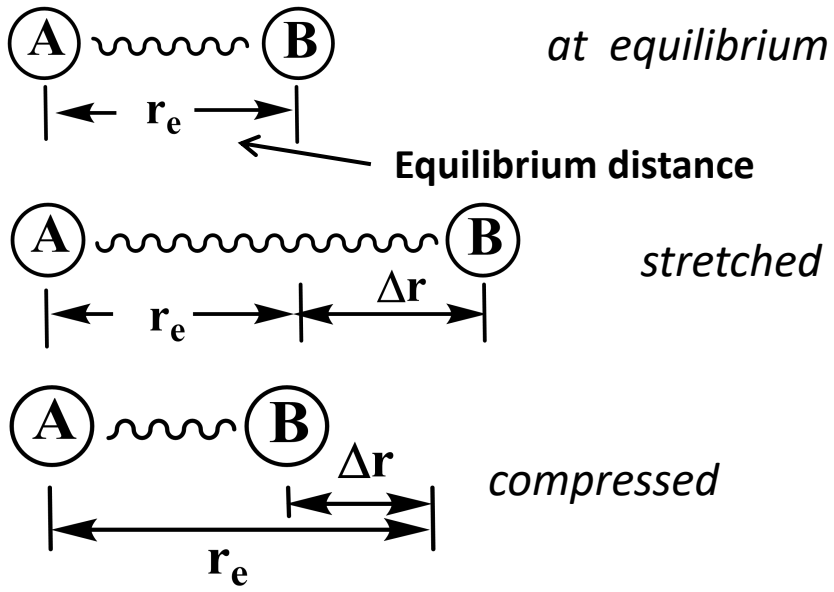
Vibrational Spectroscopy

The individual atoms of a molecule are constantly in motion are the entire range of temperatures above absolute zero.

⇩

Vibrations, translations, and rotations

Diatomic molecules:



分子的各个自由度

Stretching/compressing movements can be treated as a spring, using Hooke's law:

$$\boldsymbol{F = -\, k\Delta r}$$

Force constant

For such harmonic oscillation(obeying classical mechanics):

Potential energy:

$$\boldsymbol{V = \frac{1}{2}\kappa(\Delta r)^2}$$

a parobolic function

振动的弹簧模型

The energy of a vibrating molecule is subject to quantum mechanical restrictions:

The oscillating molecule can assume only certain levels of vibrations energy:

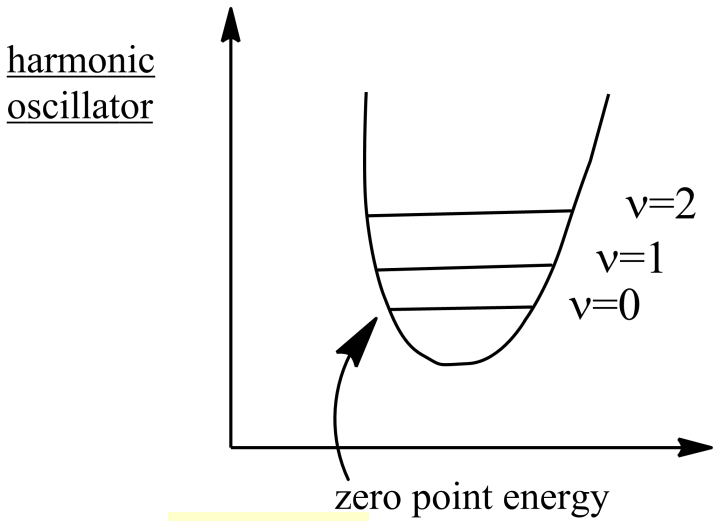
$$E(v) = h\nu\left(v + \frac{1}{2}\right)$$

Vib. frequencies Vibrational quantum#

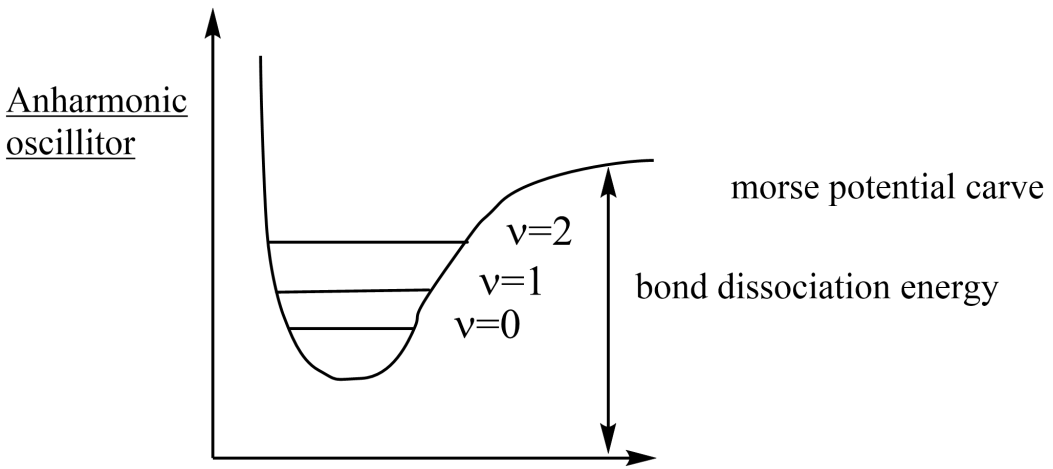
where $\nu = \frac{1}{2\pi}\sqrt{\frac{\kappa}{\mu}}$

约化质量

$$\mu = {m_A \cdot m_B}/{m_A + m_B}$$



零点能



$$E(v) = h\nu\left(v + \frac{1}{2}\right) - h\nu X_e\left(v + \frac{1}{2}\right)^2 + \cdots$$

X_e—*anharmonicity constant*

Quantum mechanical selection rule: $\Delta v = \pm 1$

IR: $v = 0 \rightarrow v = 1$

↓ ↘ 1st excited state

Vibrational g.s. (totally symmetric)

光谱选律

Raman: Probing vibrations using visible wavelength

Stokes line $v = 0 \rightarrow v = 1$

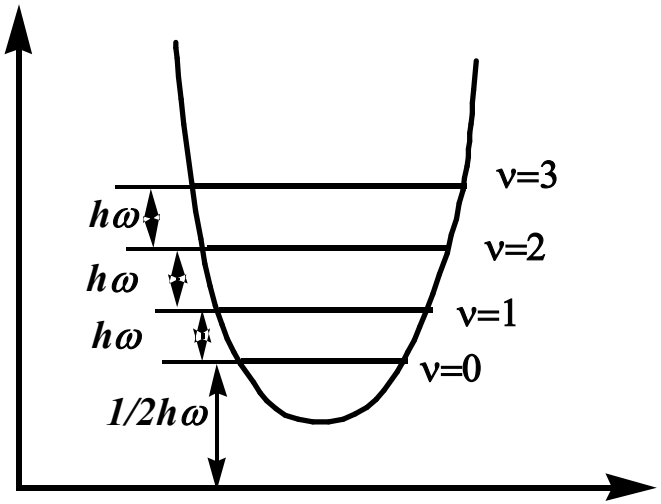
Antistokes line $v = 1 \rightarrow v = 0$

Harmonic oscillator: $E(v) = h\nu\left(v + \frac{1}{2}\right) = h\omega\left(v + \frac{1}{2}\right)$

$$\Delta E = E(v) - E(0) = h\omega \cdot \nu$$

0→1 transition :$\Delta E = h\omega$ fundamental frequency

振动光谱术语



Anharmonic oscillator:

$$V = De\left(1 - e^{-\beta q}\right)$$ The morse potential

$$\overline{E}(v) = \overline{w}_e\left(v + \frac{1}{2}\right) - \overline{w}_e X_e\left(v + \frac{1}{2}\right)^2$$

$$\Delta E_{(0\to v)} = \overline{E}_v - \overline{E}_0$$

$$= \overline{w}_e v - \overline{w}_e X_e v(v+1)$$

$v = 0 \to v = 1$ Fundamental transition

$v = 0 \to v = 2$ The first overtone

$v = 0 \to v = 3$ The second overtone

cm^{-1}

H_2:	4159.5	HI:	2230.0
D_2:	2980.3	F_2:	892
HF:	3958.4	Cl_2:	556.9
HCl:	2885.7	…	…
HBr:	2559.2	Na_2:	157.8
		KCl	278

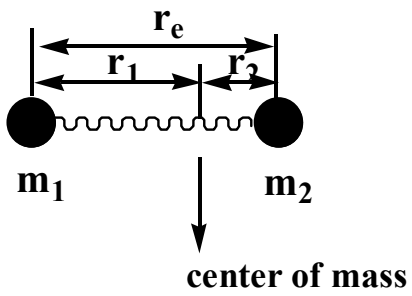
Observed and calculated vibrational energies of HCl:

	$\bar{\nu}_{obs}(cm^{-1})$	$\bar{\nu}_{calc}(harmonic)$	$\bar{\nu}_{calc}(anharmonic)$
$0 \rightarrow 1$	2885.9	2885.9	2885.7
$0 \rightarrow 2$	5668.0	5771.8	5668.2
$0 \rightarrow 3$	8347.0	8657.7	8347.5
$0 \rightarrow 4$	10,923.1	11,543.6	10,923.6
$0 \rightarrow 5$	13,396.5	14,429.5	13,396.5

Rotation:

分子的转动

The rigid rotor model



The molecule rotates about the center of mass (defined by $m_1r_1=m_2r_2$)

The moment of inertia: $I = \sum_i m_i {r_i}^2$

For diatomic molecule: $I = \frac{m_1 \cdot m_2}{m_1 + m_2} {r_e}^2 = \mu {r_e}^2$

Quantum mechanics treatment of the rigid rotor:

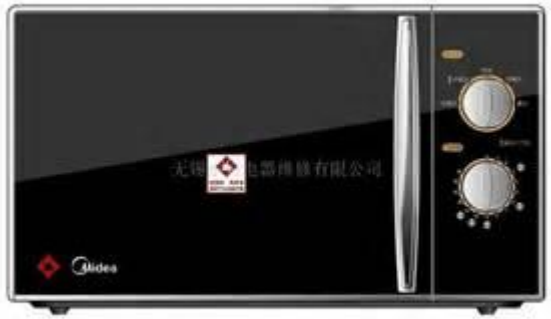
$$E_J = \frac{h^2}{2I} J(J+1) = BJ(J+1)$$

Rotational quantum number

转动光谱

$$E_J = \frac{h^2}{2I} J(J+1) = BJ(J+1)$$

For $^{12}C^{16}O$: $E_J = 1.93J(J+1)\ \text{cm}^{-1}$



For *J=1* to *J=2* transition,

$$\Delta E = 11.6 - 3.9 = 7.7\text{cm}^{-1}$$

,in the microwave portion of spectrum.

Selection rule for the rigid rotor: $J \to J+1\ or\ J-1$

$$\Delta\overline{E}(J \to J+1) = 2\overline{R}J(J+1)$$

Nonrigid rotor

$$E_J = BJ(J+1) - DJ^2(J+1)^1$$

centrifugal distortion constant

跃迁强度与什么因素相关？

Transitions between stationary states:

$\psi_0 \rightarrow \psi_1$ transition

1-D: $\psi(x,t) = c_0\psi_0(x,t) + c_1\psi_1(x,t)$

$$\frac{d({c_1}^* \cdot c_1)}{d_t} \propto {I_o}^2 {M_{01}}^2 \longrightarrow \underline{\text{Transition moment integral}}$$

跃迁偶极矩

Intensity of radiation

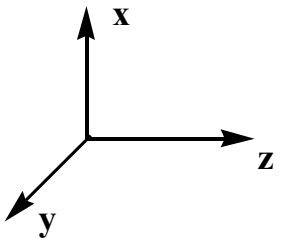
Initial rate at which the excited rate is occupied.

吸光能力

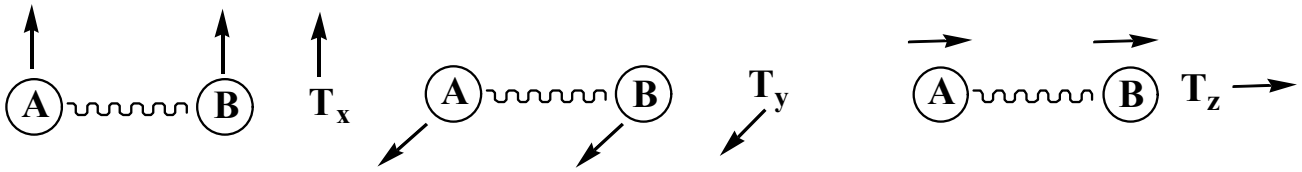
简正模式

The normal modes of vibrations of polyatomic molecules

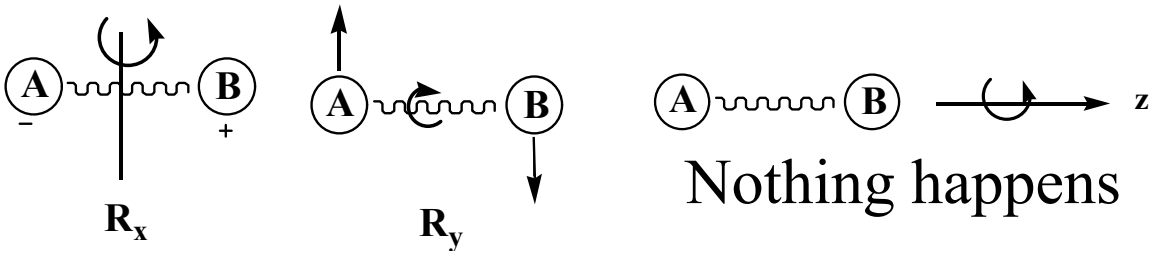
Example: linear molecule A-B:



Translation:



Rotation:

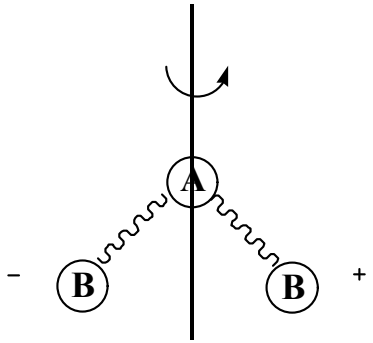


The degree of freedom for a molecule with N atoms is 3N.

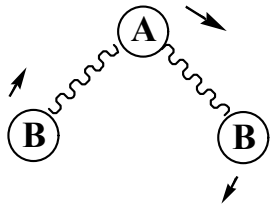
∴for linear molecules, number of vibrational modes is **3N-5**

自由度

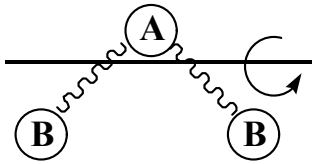
For nonlinear molecules: BAB



Rx



Ry



Rz

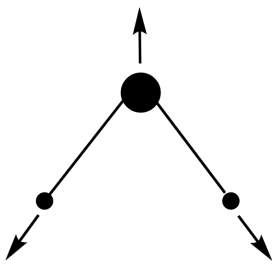
Rz

Number of vibrational modes is : 3N-6

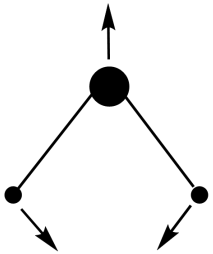
Each normal mode of vibration forms a basis for an irreducible representation of the point group of the molecule

Three normal modes of H_2O vibrations:

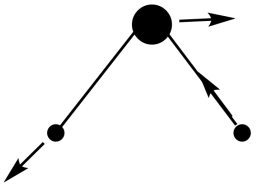
水的简正模式



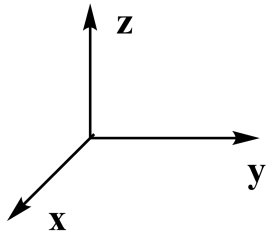
υ_1
(symmetric stretch)



υ_2
(bend)



υ_3
(antisymmetric stretch)

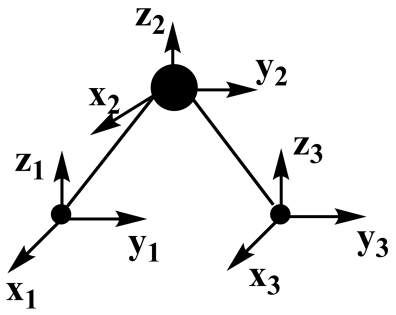


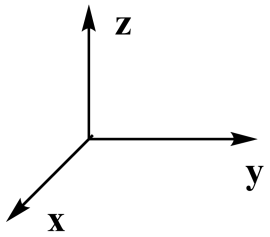
Symmetries of these normal modes:

C_{2v}	E	C_2	$\sigma_{(xz)}$	$\sigma_{(yz)}$		
ν_1	1	1	1	1	=	a_1
ν_2	1	1	1	1	=	a_1
ν_3	1	-1	-1	1	=	b_2

But we typically do not know the normal modes of vibrations for a molecule! Can we use what we have learned so far to figure out the symmetries of vibrational normal modes w/o knowing the normal modes first? **YES**

Look at H_2O again:





For E operation: $x_1 \to x_1, y_1 \to y_1, z_1 \to z_1$

$x_2 \to x_2, y_2 \to y_2, z_2 \to z_2$

In matrix form: $x_3 \to x_3, y_3 \to y_3, z_3 \to z_3$

The character of the matrix is 9

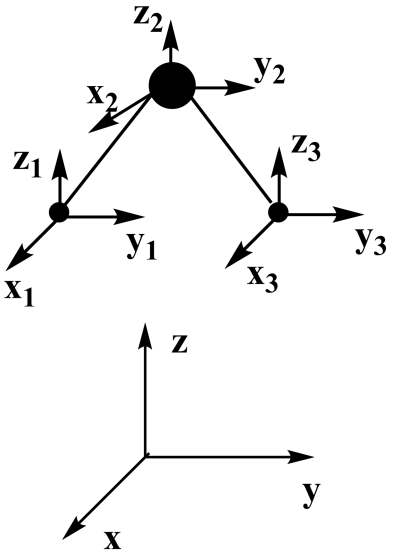
你可能不太相信，特征标表里的数字就是这类矩阵的对角线元素和
具体原因在ppt你跳过的部分，反正你也看不懂，不妨相信数学家

$$E \bullet \begin{pmatrix} x_1 \\ y_1 \\ z_1 \\ x_2 \\ y_2 \\ z_2 \\ x_3 \\ y_3 \\ z_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \bullet \begin{pmatrix} x_1 \\ y_1 \\ z_1 \\ x_2 \\ y_2 \\ z_2 \\ x_3 \\ y_3 \\ z_3 \end{pmatrix} = \begin{pmatrix} x_1 \\ y_1 \\ z_1 \\ x_2 \\ y_2 \\ z_2 \\ x_3 \\ y_3 \\ z_3 \end{pmatrix}$$

C_2 operation: $x_1 \to x_3,\quad x_2 \to -x_2,\quad x_3 \to -x_1$

$y_1 \to -y_3,\ y_2 \to -y_2,\quad y_3 \to -y_1$

$z_1 \to +z_3,\quad z_2 \to +z_2,\quad z_3 \to +z_1$



$$C_2 \bullet \begin{pmatrix} x_1 \\ y_1 \\ z_1 \\ x_2 \\ y_2 \\ z_2 \\ x_3 \\ y_3 \\ z_3 \end{pmatrix} = \begin{pmatrix} 0&0&0&0&0&0&-1&0&0 \\ 0&0&0&0&0&0&0&-1&0 \\ 0&0&0&0&0&0&0&0&+1 \\ 0&0&0&-1&0&0&0&0&0 \\ 0&0&0&0&-1&0&0&0&0 \\ 0&0&0&0&0&+1&0&0&0 \\ -1&0&0&0&0&0&0&0&0 \\ 0&-1&0&0&0&0&0&0&0 \\ 0&0&+1&0&0&0&0&0&0 \end{pmatrix} \bullet \begin{pmatrix} x_1 \\ y_1 \\ z_1 \\ x_2 \\ y_2 \\ z_2 \\ x_3 \\ y_3 \\ z_3 \end{pmatrix} = \begin{pmatrix} -x_3 \\ -y_3 \\ z_3 \\ -x_2 \\ -y_2 \\ z_2 \\ -x_1 \\ -y_1 \\ z_1 \end{pmatrix}$$

The character of C_2 matrix is **-1**

总结：被对称操作移动了位置的原子对特征标无贡献

Similarly, we can deduce that the character for $\sigma_{(xz)}$ and $\sigma_{(yz)}$ are 1 and 3, respectively.

Using the decomposition formula, we can determine:

2.分解为更基本的对称性(不可约表示)

$$P_{tot} = 3a_1 + a_2 + 2b_1 + 3b_2$$

From the character table of C_{2v}, we know:

1.把所有对称操作的矩阵都算一遍

C_{2v}	E	C_2	$\sigma_{(xz)}$	$\sigma_{(yz)}$	
A_1	1	1	1	1	z
A_2	1	1	-1	-1	R_z
B_1	1	-1	1	-1	x, R_y
B_2	1	-1	-1	1	y, R_x
P_{tot}	9	-1	1	3	

$(x,\ y,\ z)$ transform as $A_1 + B_1 + B_2 \rightarrow translation$

$(R_x,\ R_y,\ R_z)$ transform as $A_2 + B_1 + B_2 \rightarrow rotation$

The symmetries of vibrational normal modes are thus:

$$P_{tot} - P_{transtational} - P_{rotational} = 2a_1 + b_2 \quad !!$$

Streamlined procedures for determining the symmetries of vibrational normal modes:

Only the diagonal terms in the matrices contribute

简化上述过程

⇨ Atoms that move during the symmetry operation do not contribute

1) Determine the number of atoms which do not change location during each symmetry operation

2) For each operation, multiply the number of unmoved atoms by the character of $P_{x,y,z}$ at the bottom of the character table to give P_{tot}

C_{2v}	E	C_2	$\sigma_{(xz)}$	$\sigma_{(yz)}$	
$P_{x,y,z}$	3	-1	1	1	= $A_1+B_1+B_2$
#of unmoved atoms	3	1	1	3	
P_{tot}	9	-1	1	3	= $3a_1+a_2+2b_1+3b_2$

"see earlier character table"

*(get this info from character table)

$P_{x,y,z}$ 怎么来的？见后页

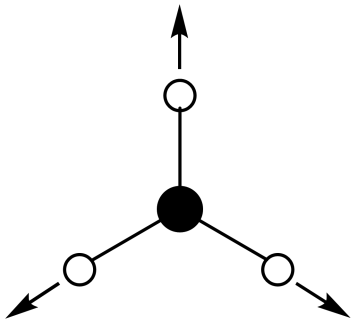
It gives the same answer but much simpler

C_{2v}	E	C_2	$\sigma_{(xz)}$	$\sigma_{(yz)}$		
A_1	1	1	1	1	z	
B_1	1	-1	1	-1	x	
B_2	1	-1	-1	1	y	
P_{xyz}	3	-1	1	1		

直接把x,y,z对应的不可约表示的特征标加起来就得到$P_{x,y,z}$

BCl_3 D_{3h}

D_{3h}	E	$2C_3$	$3C_2$	σ_h	$2S_3$	$3\sigma_v$
$P_{x,y,z}$	3	0	-1	1	-2	1
Unmoved atoms	4	1	2	4	1	2
P_{tot}	12	0	-2	4	-2	2

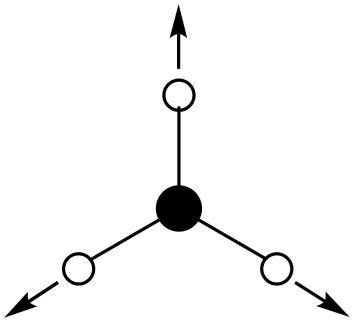


Using decomposition formula:

$$P_{tot} = A_1{}' + A_2{}' + 3E' + 2A_2{}'' + E''$$

$$P_{trans} = E' + A_2{}'' \qquad P_{rot} = A_2{}' + E''$$

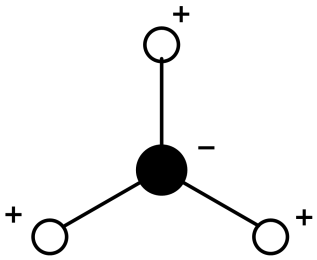
$$\therefore P_{vib} = A_1{}' + 2E' + A_2{}''$$



$\nu_1(A_1')$

$^{10}BCl_3$ 471 cm^{-1}

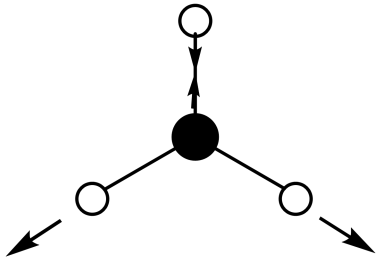
$^{11}BCl_3$ 471 cm^{-1}



$\pi(A_2'')$

$^{10}BCl_3$ 480 cm^{-1}

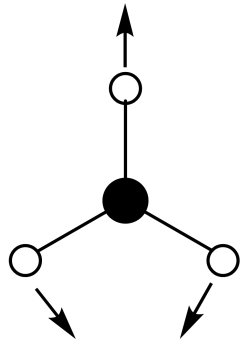
$^{11}BCl_3$ 460 cm^{-1}



$\nu_d(E')$

$^{10}BCl_3$ 995 cm^{-1}

$^{11}BCl_3$ 956 cm^{-1}



$\delta_d(E')$

$^{10}BCl_3$ 244 cm^{-1}

$^{11}BCl_3$ 243 cm^{-1}

ν—stretching

δ—deformation(bending)

ρ_w—wagging

ρ_r—rocking

ρ_t—twisting

π—out of plane

as—asymmetric

s—symmetric

d—degenerate

这些振动模式中哪些是红外活性的呢？

Selection Rule

跃迁的选律

We showed earlier transition between stationary states $\psi_0 \rightarrow \psi_1$ can be described as :

$$transition\ probability = \frac{d(c_1{}^* \cdot c_1)}{dt} \propto \varepsilon_0{}^2 \cdot M_{01}{}^2$$

Transition moment integral

For a one-dimensional systems the IR transition moment integral is:

$$M_{vv'} = \int_{-\infty}^{\infty} \psi^*(v') \cdot \mu \cdot \psi(v) \cdot dx$$

积分是否为0？是奇函数还是偶函数？

We need to determine if this integral is zero or non-zero to figure out if the transition is allowed or forbidden

For a 3-D system: transition probability $\propto M_{vv'}{}^2 = \left[\int_{-\infty}^{\infty} \psi^*(v') \cdot \hat{\mu}\psi(v) dI\right]$

$\hat{\mu}$ transforms as (x, y, and z)

IR Selection Rule: If the direct product $P_{[\psi(v')]} \times P_{(\hat{\mu})} \times P_{[\psi(v)]}$ contains the totally symmetric irreducible representation of the point group, the transition $v \rightarrow v'$ is IR active.

这么多振动模式中哪几个是红外活性的呢？

例 $A_1{}'$ For the fundamental transition $v = 0 \rightarrow v = 1$

$$M_{01} = \int_{-\infty}^{\infty} \psi(1) \cdot \hat{\mu} \cdot \psi(0) \cdot dI$$

The symmetry of the integrand is:

$\hat{\mu}$ transforms as (x, y, and z)

$$\psi(1)\hat{\mu}\psi(0) \sim A_1{}' \begin{pmatrix} A_2{}'' \\ E' \end{pmatrix} A_1{}' = \begin{pmatrix} A_2{}'' \\ E' \end{pmatrix}$$

Means "transforms as"

见下页的定理和推论

$\psi(0)$ is always totally symmetric

$\psi(1)$ has the symmetry of the normal vibration($A_1{}'$in this case)

Since the direct product $P_{[\psi(1)]} \cdot P_{(\hat{\mu})} \cdot P_{[\psi(0)]}$ does not contain totally symmetric irreducible representation, this fundamental transition is not allowed!

$\psi(0)$ is always totally symmetric

定理一：每个点群都有一个不可约表示是一维全对称的，也就是说在所有对称操作下保持不变，特征标全为1

推论一：基态波函数的对称性永远属于全对称不可约表示。

因为点群的判定是基于基态分子对称性的，如果基态波函数在某个点群的对称操作下不保持不变，那么这个分子必然不是这个点群的。

定理二：任何不可约表示与全对称不可约表示的乘积等于自己

全对称相当于1

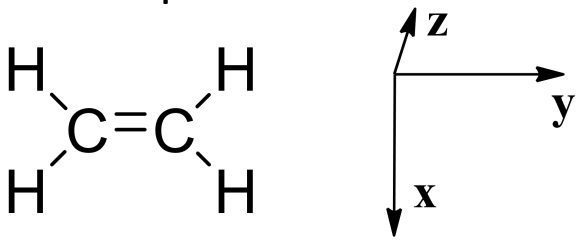
定理三：只有对称性为（包含）全对称不可约表示的函数，在全空间的积分才不为零

全对称相当于一维积分里的偶函数，其他对称性都相当于奇函数

定理四：两个不同不可约表示的乘积必然不包含全对称不可约表示；两个相同不可约表示的乘积才包含全对称不可约表示。

红外跃迁的选律

Look at ethene as an example:



D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$	
$P_{x,y,z}$	3	-1	-1	-1	-3	1	1	1	
Unmoved atoms	6	0	2	0	0	6	0	2	
P_{tot}	18	0	-2	0	0	6	0	2	
P_{trans}	3	-1	-1	-1	-3	1	1	1	$= B_{1u} + B_{2u} + B_{3u}$
P_{rot}	3	-1	-1	-1	3	-1	-1	-1	$= B_{1g} + B_{2g} + B_{3g}$
$P_{vib} = P_{tot} - P_{trans} - P_{rot}$	12	2	0	2	0	6	0	2	

$$= 3A_g + 2B_{1g} + B_{3g} + A_u + B_{1u} + 2B_{2u} + 2B_{3u}$$

$B_{1u} + 2B_{2u} + 2B_{3u}$ 都是红外活性的，这个分子有5个红外活性的振动模式

Raman Selection Rule: If the direct product $P_{[\psi(v')]} \times P_{(\hat{\alpha})} \times P_{[\psi(v)]}$ contains the totally symmetric irreducible representation of the point group, the transition $v \rightarrow v'$ is Raman active.

For Raman transitions, $\hat{0}$ will be the polarizability operator $\hat{\alpha}$. $\hat{\alpha}$ has six different component(α_{xx}, α_{yy}, α_{zz}, α_{xy}, α_{yz}, α_{xz}). These six components of α transform as the binary products x^2, y^2, z^2, xy, xz, and yz.

$$M_{01} = \int_{-\infty}^{\infty} \psi(1) \cdot \hat{\alpha} \cdot \psi(0) \cdot dI$$

拉曼的选律

For ethene $\hat{\alpha} \sim \begin{pmatrix} x^2, y^2, z^2 \sim A_g \\ xy \sim B_{1g} \\ xz \sim B_{2g} \\ yz \sim B_{3g} \end{pmatrix}$

$$\boldsymbol{3A_g + 2B_{1g} + B_{3g}}$$

这6个振动模式是拉曼活性的

Simplified Selection Rules: for fundamental transitions, the only IR-active modes will have the symmetries of the three components of $\hat{\mu}$. The only Raman –active modes will have the symmetries of the six components of $\hat{\alpha}$.

XeF_4: $P_{vib} = a_{1g} + b_{1g} + b_{2g} + a_{2u} + b_{2u} + 2e_u$

中心对称分子没有一个振动模式是同时有红外和拉曼活性的

$\hat{\mu} \sim a_{2u} + e_u$

$\hat{\alpha} \sim a_{1g} + b_{1g} + b_{2g} + e_g$

$\therefore IR - active: \ a_{2u} + 2e_u$

$Raman - active: \ a_{1g} + b_{1g} + b_{2g}$

Mutual exclusion rule for molecules w/. A center of symmetry:

$\hat{\mu}$ will always have u symmetry and $\hat{\alpha}$ will always have g symmetry.

No mode can be both IR-and Raman-active for a centrosymmetric molecule.

If strong bands are found at the same frequency for both the IR and Raman spectra, it is likely that the molecules does not have a center of symmetry.

Polarization:

For H_2O molecules, $\boldsymbol{P_{vib}} = \underset{v_1+v_2}{\boldsymbol{2a_1}} + \underset{v_3}{\boldsymbol{b_2}}$

光谱的偏振

$$\hat{\boldsymbol{\mu}} \sim \underset{z}{\boldsymbol{a_1}} + \underset{x}{\boldsymbol{b_1}} + \underset{y}{\boldsymbol{b_2}}$$

$$v_1 \ or \ v_2: \quad \psi(1)\hat{\mu}\psi(0) \sim a_1 \begin{pmatrix} a_1 \\ b_1 \\ b_2 \end{pmatrix} a_1 = \begin{pmatrix} a_1 \\ b_1 \\ b_2 \end{pmatrix}$$

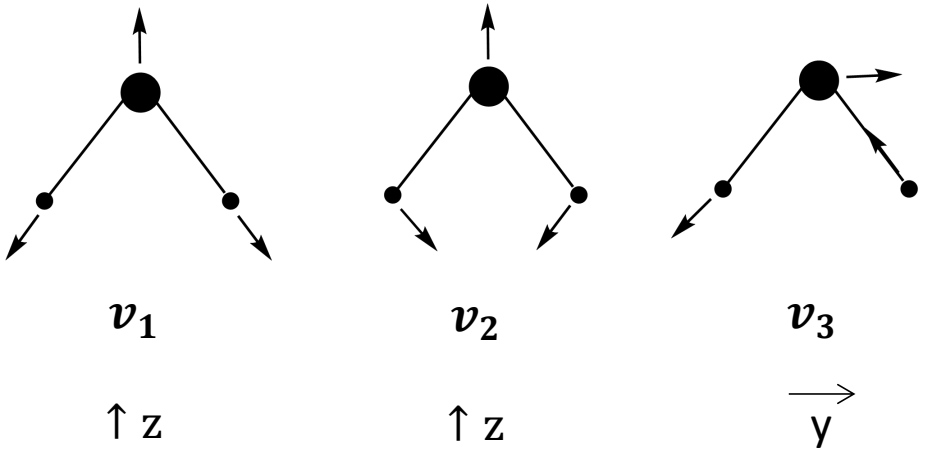
$$v_3: \quad \psi(1)\hat{\mu}\psi(0) \sim b_2 \begin{pmatrix} a_1 \\ b_1 \\ b_2 \end{pmatrix} a_1 = \begin{pmatrix} b_2 \\ a_2 \\ a_1 \end{pmatrix}$$

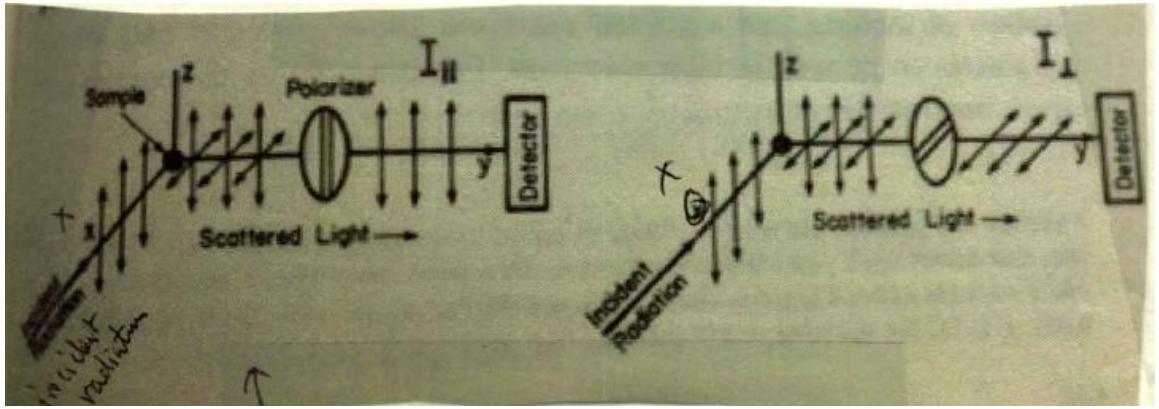
$\therefore$ *All three transitions are IR*

$-$ *allowed and show up at* $3657\ cm^{-1}$, $1595\ cm^{-1}$, *and* $3756\ cm^{-1}$, *respectively*

只有电场沿z方向才能激发$\boldsymbol{v_1}$和$\boldsymbol{v_2}$

只有电场沿y方向才能激发$\boldsymbol{v_3}$





Solution sample

The intensity of scattered light polarized in the same plane as the incident light is $I_{\parallel}$, while that of light polarized perpendicular to the incident beam is $I_{\perp}$.

Depolarization ratio $\boldsymbol{\rho = {I_{\perp}}/{I_{\parallel}}}$

For a solution sample, $\rho = {3}/{4}$ for a depolarized(unpolarized) band

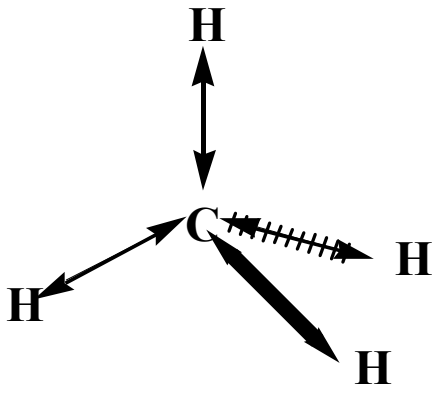
$0 \leq \rho \leq \frac{3}{4}$ for a polarized band, which can only arise from vibrations that are totally symmetric.

$\rho \rightarrow$ closer to zero for the totally symmetric vibrations of highly symmetrical molecules.

Totally symmetric vibrations give the strongest Raman bands.

Stretching mode analysis:

伸缩振动分析



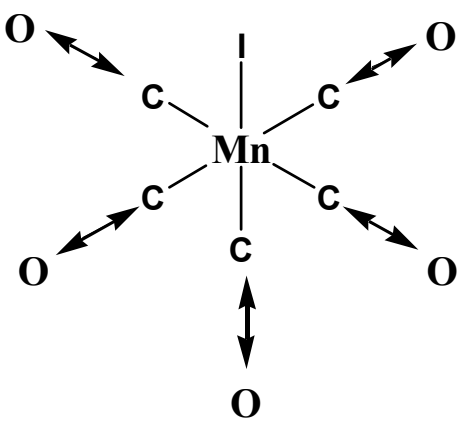
T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$	
$P_{stretch}$	4	1	0	0	2	$= a_1 + t_2$

$P_{(xy,\ xz,\ yz)} = t_2$ $\qquad$ $P_{(square\ terms)} = a_1 + e$

$P_{(x,y,z)} = t_2$

t_2 stretching mode is IR-active
$a_1 + t_2$ stretching mode are both Raman-active

Carbonyl stretching bands in $Mo(CO)_5I$:



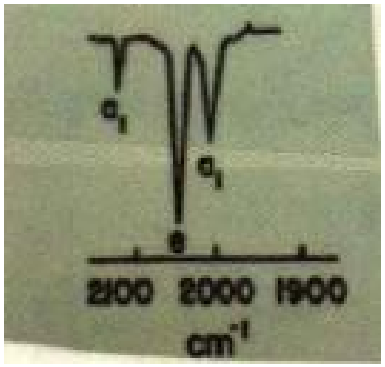
C_{4v}	E	$2C_4$	C_2	$2\sigma_v$	$2\sigma_d$ ($\sigma_{v'}$)
$P_{stretch}$	5	1	1	3	1

$$= 2a_1 + b_1 + e$$

$z \sim a_1$ $\quad (x, y) \sim e$

$x^2 + y^2,\ z^2 \sim a_1$ $\quad x^2 - y^2 \sim b_1,$

$xy \sim b_2$ $\quad (xz,\ yz) \sim e$

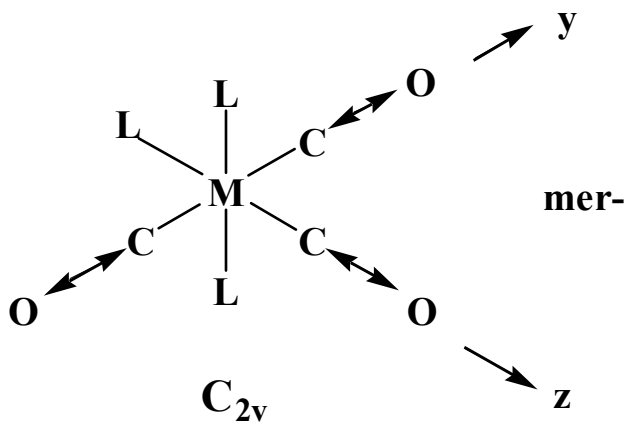


(IR spectrum in CCl_4)

IR-active: $2a_1 + e$

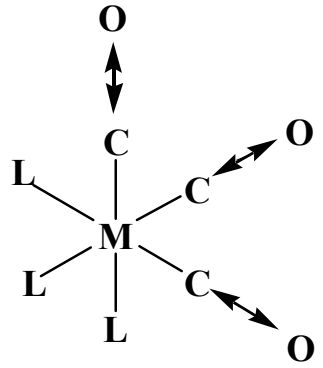
Raman: $2a_1 + b_1 + e$

Can we differentiate mer- and fac- isomers of $ML_3(CO)_3$



mer-

C_{2v}



fac-

C_{3v}

C_{2v}	E	C_2	$\sigma_{(xz)}$	$\sigma_{(yz)}$
$P_{stretch}$	3	1	1	3

$$= 2a_1 + b_2$$

All three ir-active

C_{3v}	E	$2C_3$	$3\sigma_V$
$P_{stretch}$	3	0	1

$$= a_1 + e$$

both ir-active

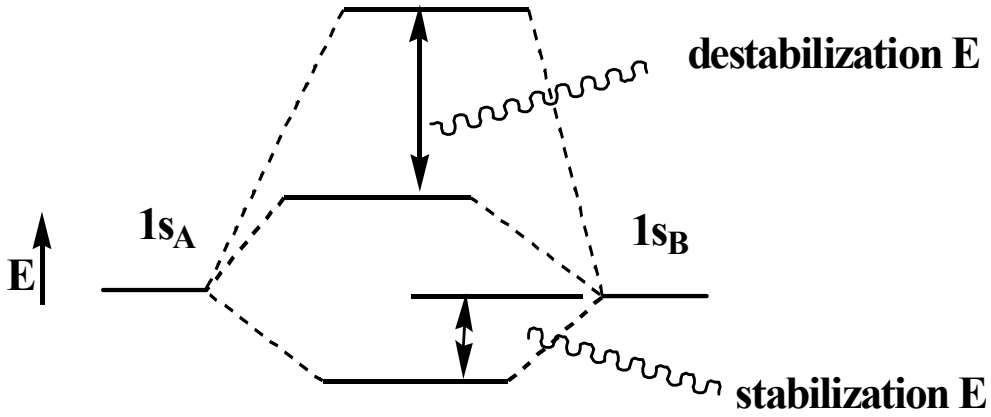
$Mo(CO)_3[P(OCH_3)_3]_3$: IR spectrum 1993, 1919, 1890 $cm^{-1} \rightarrow mer-$

$Cr(CO)_3(CNCH_3)$: IR spectrum 1942, 1860 $cm^{-1} \rightarrow fac-$

第七节、从原子轨道组成分子轨道的对称性要求

目标：描述分子轨道在对称操作下变换的对称性

For heterodiatomic molecules:



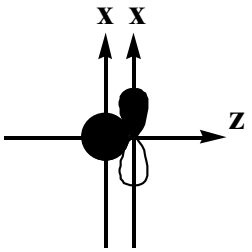
Different contributions to ψ_b and ψ^*

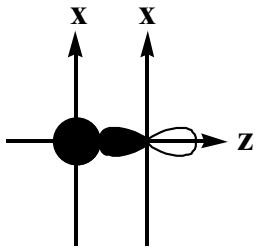


Constructing molecular orbitals:

1) In order for two orbitals to form a molecular orbital, they must have the same symmetry about the inter-nuclear axis
2) Combining orbitals of very different energies leads to only small interactions
3) Each possible molecular orbitals must form a basis for some irreducible representation of the point group of the molecular

$S_{sp_x} = 0$





$S_{sp_z} \neq 0$

交换积分

$$H_{ab} = \int \psi_a^{1s^*} \hat{H} \psi_b^{1s} dv$$

$$\int \psi_a^{1s^*} \psi_b^{1s} dv$$

（重叠积分S_{ab}）

Let's look at homo diatomic molecules again (s orbitals):

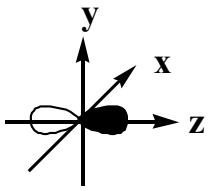
$\begin{pmatrix} 1s_A + 1s_B \\ 1s_A - 1s_B \end{pmatrix}$ ⇦ Variation method

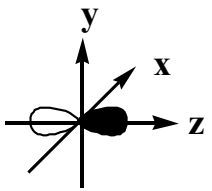
$\mathbf{D_{\infty h}}$	E	$\mathbf{2C_\infty^\phi}$	$\mathbf{\infty\sigma_v}$	i	$\mathbf{2S_\infty^\phi}$	$\mathbf{\infty C_2}$	
$\boldsymbol{P_{(1s)}}$	**2**	**2**	**2**	**0**	**0**	**0**	$= \boldsymbol{\sigma_g{}^+ + \sigma_u{}^+}$

For bonding and antibonding MO's derived from variation method

$\mathbf{D_{\infty h}}$	E	$\mathbf{2C_\infty^\phi}$	$\mathbf{\infty\sigma_v}$	i	$\mathbf{2S_\infty^\phi}$	$\mathbf{\infty C_2}$	
$\boldsymbol{\psi_b}$	**1**	**1**	**1**	**1**	**1**	**1**	$= \boldsymbol{\sigma_g{}^+}$
$\boldsymbol{\psi^*}$	**1**	**1**	**1**	**-1**	**-1**	**-1**	$= \boldsymbol{\sigma_u{}^+}$

How about diatomic molecules with 2s+2p orbitals:



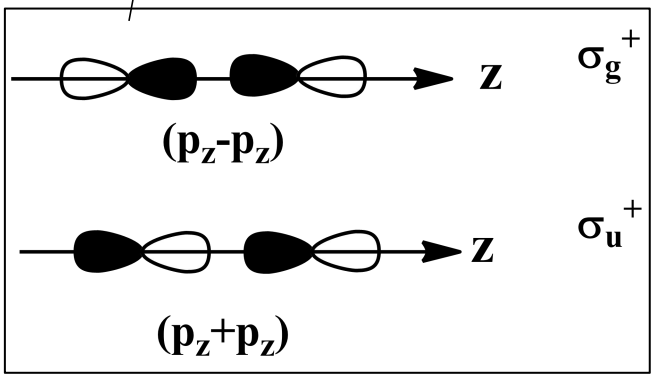


$p_z - p_z$ interactions are actually similar to 1s-1s interactions in symmetry.

$D_{\infty h}$	E	$2C_\infty^\phi$	$\infty\sigma_v$	i	$2S_\infty^\phi$	∞C_2	
$P_{(2s,\,2p_z)}$	4	4	4	0	0	0	$= 2\sigma_g^+ + 2\sigma_u^+$
$P_{(2p_x,\,2p_y)}$*	4	$4\cos\phi$	0	0	0	0	$= \pi_g + \pi_u$

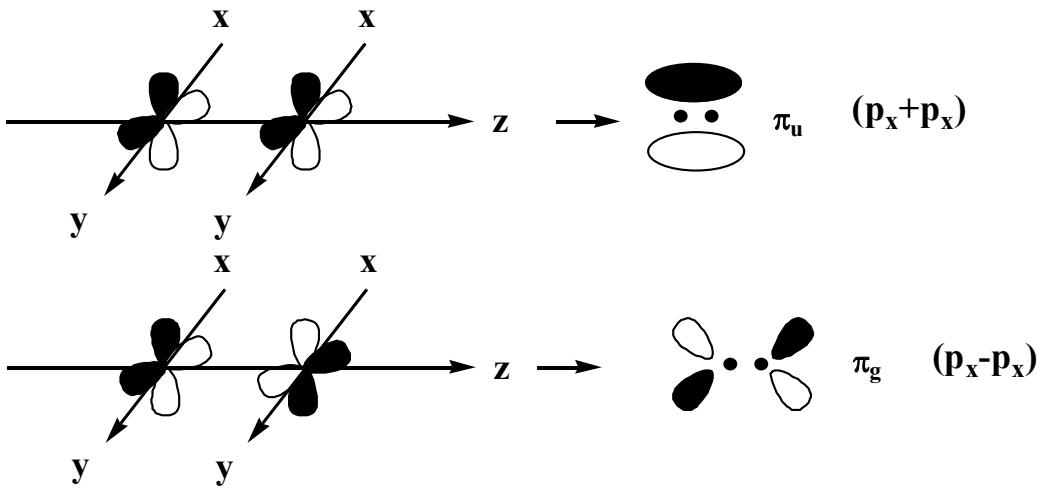
$$R_\phi \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \cos\phi & \sin\phi \\ -\sin\phi & \cos\phi \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

px, py opposite values cancel each other



$(p_x + p_x)$ and $(p_y + p_y)$ together form a basis for the doubly degenerate representation π_u

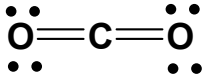
$(p_x - p_x)$ and $(p_y - p_y)$ together transforms a π_g



$(p_y + p_y)$ similar to $(p_x + p_x)$
but orthogonal.

Polyatomic Molecules:

Determine how the basis atomic orbitals tramsform to construct linear combinations of appropriate symmetries

CO_2: $D_{\infty h}$ Use the lewis structure as a guide 

* Implying a single bond and a double bond are needed form the C=O double bond

Consider 2s+2p valence orbitals:

$\mathbf{D_{\infty h}}$	E	$2C_\infty^\phi$	$\infty\sigma_v$	i	$2S_\infty^\phi$	∞C_2	
$C(2s)$	1	1	1	1	1	1	$= \sigma_g^+$
$C(2p_Z)$	1	1	1	-1	-1	-1	$= \sigma_u^+$
$C(2p_x, 2p_y)$	2	$2c\phi$	0	-2	$2c\phi$	0	$= \pi_u$
$2O(2s)$	2	2	2	0	0	0	$= \sigma_g^+ + \sigma_u^+$
$2O(2p_Z)$	2	2	2	0	0	0	$= \sigma_g^+ + \sigma_u^+$
$2O(2p_x, 2p_y)$	4	$4c\phi$	0	0	0	0	$= \pi_g + \pi_u$

Polyatomic Molecules:

具无穷次轴点群的特征标分析

先看无穷次轴的特征标，

$$\chi(C\infty) = a_0 + 2a_1 cos\varphi + 2a_2 cos2\varphi$$

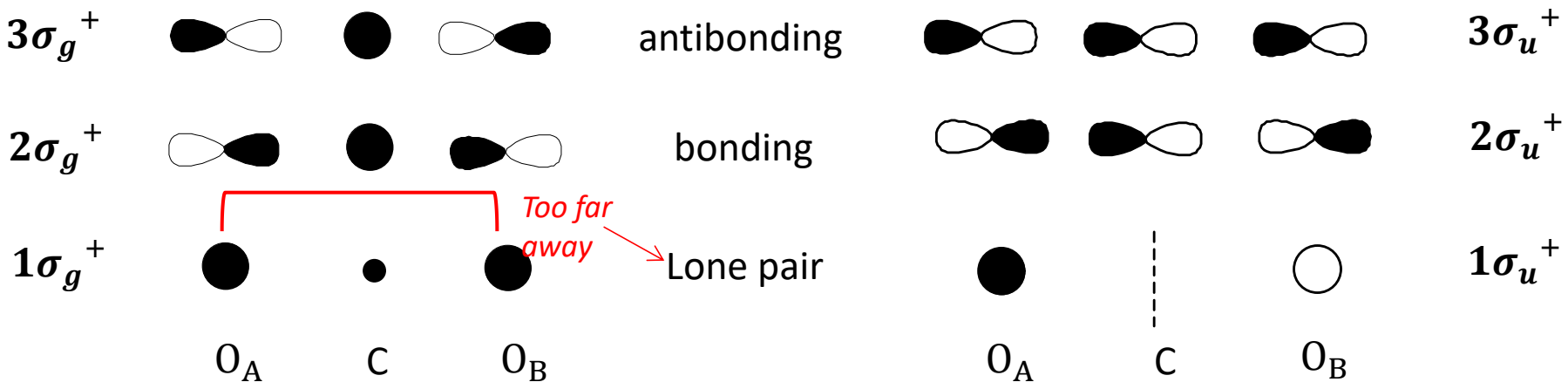
则
a_0 是所有Σ对称性数目的和
a_1 是所有Π对称性数目的和
a_2 是所有Δ对称性数目的和

如果该群包含Σ^+和Σ^- 则还要考虑$\sigma_{\infty v}$的特征标

$a_0^+ = \frac{a_0 + \chi(\sigma_{\infty v})}{2}$ 是Σ^+的数目

$a_0^- = \frac{a_0 - \chi(\sigma_{\infty v})}{2}$ 是Σ^-的数目

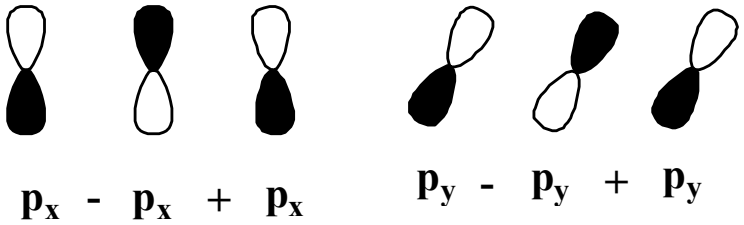
Based on the above table, C(2s), O(2s), and O($2p_Z$) will to the LCAO's of σ_g^+ symmetry.



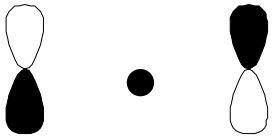
*C(2s) has much higher energy than O(2s)

You only use each orbital once!

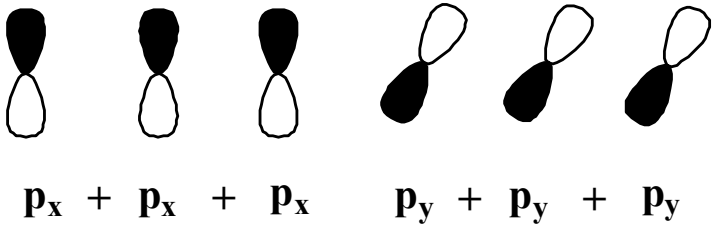
Based on the table, we need to construt two degenerate pairs of π orbitals (4 total) of π_u symmetry using $C(2p_x, 2p_y)$ & $O(2p_x, 2p_y)$ orbitals .



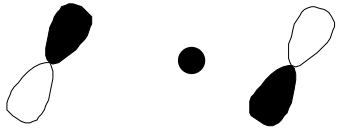
$1\pi_u$(antibonding)



$1\pi_g$(nonbonding)



$1\pi_u$(bonding)



C(2p) orbitals have u symmetry and have to have zero contribution

Symmetries of the valence A.O.S of HCN

$C_{\infty v}$	E	$2C_{\infty}^{\phi}$	$\infty\sigma_v$	
H1s + C2s + N2s	**3**	**3**	**3**	$= 3\sigma_g^+$
$C2p_Z$+$N2p_Z$	**2**	**2**	**2**	$= 2\sigma_g^+$
$C(2p_x, 2p_y) + N(2p_x, 2p_y)$	**4**	**4c ϕ**	**0**	$= \pi$

LCAO=linear combination of atomic orbitals

MO diagram for methane:

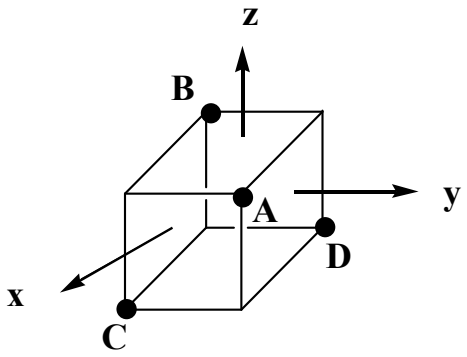
C(1s) very low in energy → nonbonding

The delocalized MO's formed by **4 H(1s) + C(2s, 2p**) orbitals

4H atoms → a set of tetrahedrally oriental vectors pointing toward He central carbon atom:

Symmetry-adapted linear combinations of atormic orbitals (SALCs)

T_d	E	$8C_3$	$3C_2$	$6S_4$	$6\sigma_d$	
P_{SALC}	4	1	0	0	2	$= a_1 + t_2$



$$\phi_1 = \frac{1}{2}[1S_A + 1S_B + 1S_C + 1S_D] \qquad a_1 \text{ symmetry}$$

$$\phi_2 = \frac{1}{2}[1S_A + 1S_B - 1S_C - 1S_D]$$

$$\phi_3 = \frac{1}{2}[1S_A - 1S_B - 1S_C + 1S_D] \qquad t_1 \text{ symmetry}$$

$$\phi_4 = \frac{1}{2}[1S_A - 1S_B + 1S_C - 1S_D]$$

On carbon: both 1s and 2s orbitals have a_1 symmetry, so 2s orbirals will form bonding and antibonding orbitals with ϕ_1:

$$\sigma_2 = c_1(2s) + c_2\phi_1 \qquad \rightarrow (\sigma_1 \text{ is nonbonding c(1s)orbital})$$

$$\sigma_6{}^* = c_3(2s) - c_4\phi_1$$

$c(2p)$ orbitals have t_2 symmetry.

$$\sigma_3 = c_5(2p_z) + c_6\phi_2$$

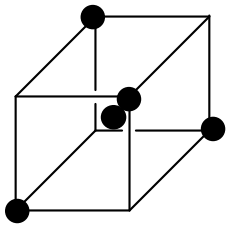
$$\sigma_7{}^* = c_7(2p_z) - c_8\phi_2$$

$$\sigma_4 = c_9(2p_y) + c_{10}\phi_3$$

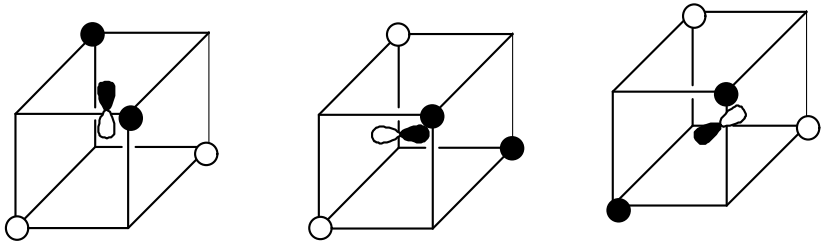
$$\sigma_8{}^* = c_{11}(2p_y) - c_{12}\phi_3$$

$$\sigma_5 = c_{13}(2p_x) + c_{14}\phi_4$$

$$\sigma_9{}^* = c_{15}(2p_x) - c_{16}\phi_4$$



$\boldsymbol{\sigma_2(a_1)}$



$\boldsymbol{\sigma_3}$ $\boldsymbol{\sigma_4}$ $\boldsymbol{\sigma_5}$

$\boldsymbol{t_2}$

直积与直和

一、直积

1. 直积的定义

两个方阵相乘可以按两种方法进行：

(1) A和B按一般矩阵的乘法规则相乘得到方阵C，称C是AB的**内积**。要求A的列数必须等于B的行数！

例：两个2×2阶矩阵相乘，得到的仍是2×2阶矩阵

$$A=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad B=\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \quad AB=C=[c_{ij}]=\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}$$

$$\begin{aligned} c_{11} &= a_{11}b_{11}+a_{12}b_{12} \\ c_{12} &= a_{11}b_{12}+a_{12}b_{22} \\ c_{21} &= a_{21}b_{11}+a_{22}b_{21} \\ c_{22} &= a_{21}b_{12}+a_{22}b_{22} \end{aligned}$$

(2) A和B可以“直接相乘”，$C=A\otimes B$称为A矩阵和B矩阵的直接乘积，简称为**直积**。对两个矩阵的阶次没有限制！

直积一般不满足交换律，即$A\otimes B\neq B\otimes A$。

“直接相乘”举例：两个2×2阶矩阵相乘，得到的是4×4阶矩阵

$$\text{假设 } A=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \qquad B=\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \qquad a_{11}+a_{22}=\chi_1;\quad b_{11}+b_{22}=\chi_2$$

$$A\otimes B=\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\otimes B=\left|\begin{array}{cc} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{array}\right|=C$$

$$C=\begin{bmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{bmatrix}$$

$$\chi_{12}=a_{11}b_{11}+a_{11}b_{22}+a_{22}b_{11}+a_{22}b_{22}$$

$$=(a_{11}+a_{22})(b_{11}+b_{22})=\chi_1\chi_2$$

2. 直积表示的特征标(character)

Taking the sum along the diagonal of each transformation matrix, known as the trace or character of the matrix

上述$A \otimes B$直积表示的特征标为：

$$\chi_{12} = a_{11}b_{11} + a_{11}b_{22} + a_{22}b_{11} + a_{22}b_{22}$$

$$= (a_{11} + a_{22})(b_{11} + b_{22}) = \chi_1\chi_2$$

说明直积方阵的迹等于单个方阵的迹的乘积。

这一结论可以归纳成量子化学中应用到的一条有关特征标方面的重要定理：

两个直积表示的特征标等于两个表示的特征标的乘积。

欲得到各属于某一个不可约表示的两种基的直积，只需将其所对应的不可约表示的特征标逐一相乘，最后将所得可约表示约化为不可约表示的直和(用符号⊕表示)。

以C_{3v}群为例

C_{3v}	E	$2C_2$	$3\sigma_v$	基
A_1	1	1	1	z, z^2
A_2	1	1	−1	
E	2	−1	0	(x, y), (xz, yz)
$\Gamma_{A1}\otimes A1$	1	1	1	
$\Gamma_{A1}\otimes A2$	1	1	−1	
$\Gamma_{A2}\otimes A2$	1	1	1	
$\Gamma_{A1}\otimes E$	2	−1	0	
$\Gamma_{A2}\otimes E$	2	−1	0	
$\Gamma_{E}\otimes E$	**4**	**1**	**0**	

约化的结果：$A_1\otimes A_1=A_1$，$A_1\otimes A_2=A_2$，$A_2\otimes A_2=A_1$，

$A_1\otimes E=E$，$A_2\otimes E=E$，$E\otimes E=A_1\oplus A_2\oplus E$

也可应用约化公式